Adaptive AI for Pipeline Counting

Capstone Project

Team Members

Total: 5 members

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Introduction

Since the company needs to count the product stocks before shipping out. There is no automatic system for counting the product. The **problems** that they found are

1. Mistake on counting by eyes always happens
    - This leads to confusion and the verification that needs to be repeated.
2. Delay in entering data into the system.
    - Since there is currently no system that the workers can enter the data directly to the main system nowadays.
3. Misrepresenting information before the data is uploaded to the system.
    - According to the previous problems, this can lead to errors when data is entered into the system later.

We decided to create a **"Product Detection Application"** which our product is Steel Pipe (in circle shape).

Why do we decide to do this project?

From our view, we want the product to be applied in a Real-World Application based on our knowledge from what we have learned, but we still want to learn more from what we have.

Since currently, we still haven't decided which way to work on for our core project, which is the object detection. These are 2 main choices that we have at the moment.

1. **OpenCV**

    Pros:
    - No model training required
    - Lightweight and fast on CPU
    - Well-supported on both iOS and Android(for app development)

    Cons:
    - We need to get the exact value for the variables that can be applied in any situation(ex. Canny, HoughCircles) because it is sensitive to lighting changes, shadows.
    - Limited accuracy in real-world scenes
2. **YOLO / AI Model**

    Pros:
    - Higher Accuracy because it handles variation in lighting, orientation and background.
    - Good on working in real-world
    - Accepted in inspection and mobile AI

    Cons:
    - Requires a lot of time to collect data and train model

- Higher computational cost (we may need the higher spec for notebook/computer on training the model).

Methodology

- Deciding on which method should be used.
- Collect some data based on the real product for test
- Creating the prototype website (for checking the result)
  In case working on **OpenCV**:
  - Making the code and tuning the variable to find the best value(based on the collected data).
  - Test with the real product

    In case that we test with the real product does not OK:
    - Find Plug-in libraries about (lightning/colors) and adapt with the code.
    - (Maybe) Find the model and apply with main.

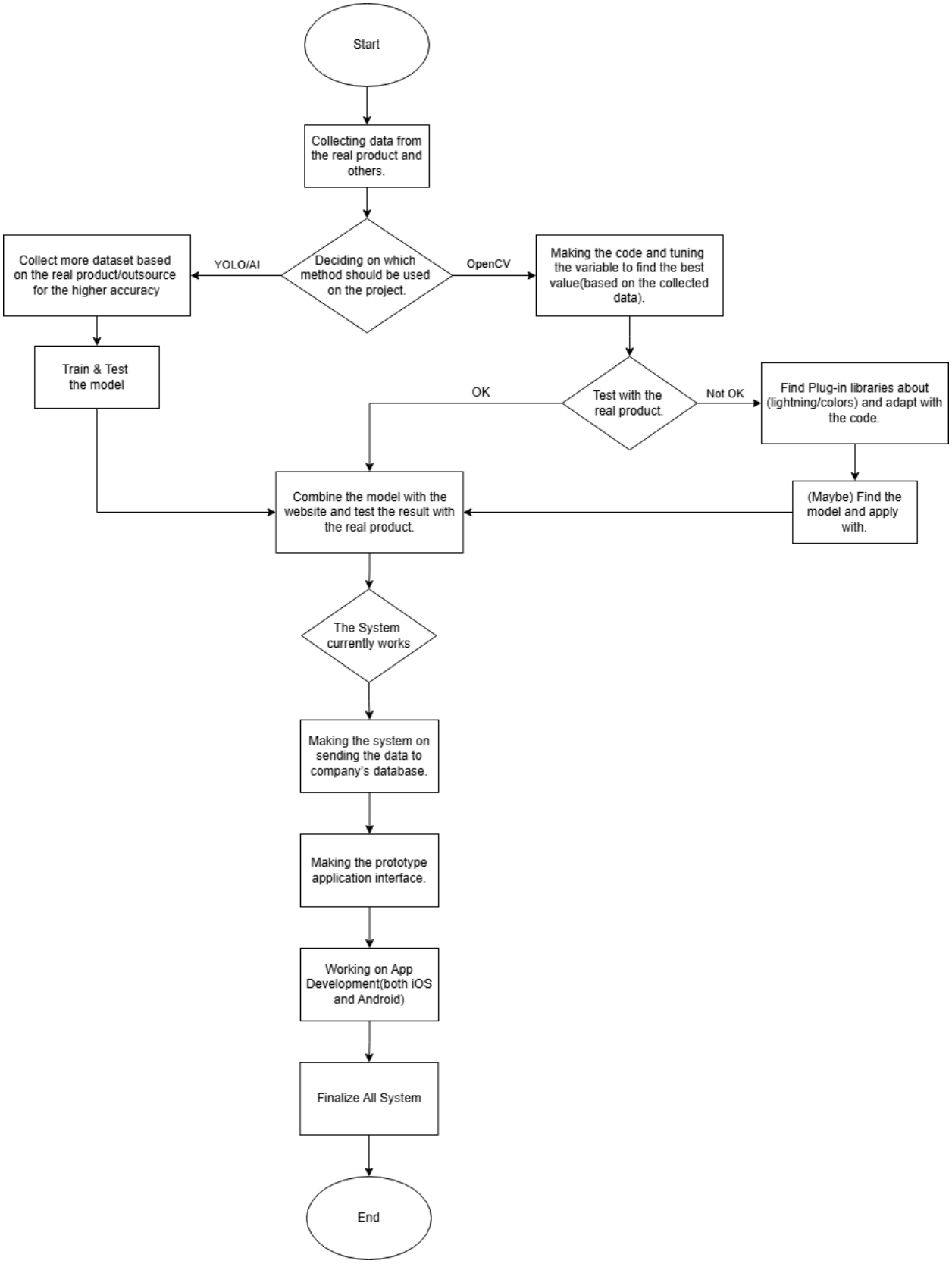
  In case working on **YOLO/AI**:
  - Collect more dataset based on the real product/outsource for the higher accuracy
  - Train & Test the model

- Combine the model with the website and test the result with the real product.

In case it currently works:

- Making the system on sending the data to company's database.
- Making the prototype application interface.
- Working on app development (both iOS and Android)
- Finalize every system

Flowchart



Mobile Application Development

Development Approach

- Cross-platform mobile application
- Single codebase supporting **Android and iOS**
- Developed using **React Native** (MIT License, open-source)

Core Functionalities

- Capture/upload images of steel pipe products using device camera
- Display detection and counting results on the mobile interface
- Allow users to confirm and submit results
- Transmit data directly to the backend server

Backend Communication

- Communication via RESTful APIs (JSON format)
- Secure data transfer using HTTPS

Why use **React Native**?

- Cross-platform development using a single codebase for **Android and iOS**
- Near-native performance through native UI components
- Reduced development time and maintenance effort
- Open-source and free to use
- Widely adopted in industry and suitable for real-world applications

Project Scope

- Being an application which can be used for object detection with high accuracy
- The application is able to send the collected data to the company's database directly.
- The application can be used on both of iOS & Android

Workload

	MLData Collect	Model Training	Object detection Code	Frontend	Backend	App dev
Game						
Hymn						
Tang						
Opor						
Guy						

	Head
	Sub

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